Lecture 13 - X_t in terms of R_j $(j \neq 0)$

Consider $P_j \cos w_j t + Q_j \sin w_j t$

Let $R_j = \frac{1}{2}(P_j - i Q_j)$ $\quad (i = \sqrt{-1})$

$$P_j \cos w_j t + Q_j \sin w_j t$$

$$= \frac{1}{2}(P_j - i Q_j)[\cos w_j t + i \sin w_j t]$$

$$+ \frac{1}{2}(P_j + i Q_j)[\cos w_j t - i \sin w_j t]$$

$$= R_j[\cos w_j t + i \sin w_j t]$$

$$+ \bar{R}_j[\cos w_j t - i \sin w_j t]$$

$$= R_j e^{i w_j t} + \bar{R}_j e^{-i w_j t}$$

$$X_t = \sum_{j=0}^{\infty} P_j \cos w_j t + Q_j \sin w_j t$$

$$= P_0 + \sum_{j=1}^{\infty} R_j e^{i w_j t} + \sum_{j=1}^{\infty} \bar{R}_j \cancel{\cos w_j t}\, e^{-i w_j t}$$